

Basic OOP Principles Theory and Reasoning

Instructions: Answer the following questions in your own words. Explain your reasoning when a situation is presented.

1. **Why are Encapsulation, Inheritance, Polymorphism, and Abstraction considered important principles of OOP? Explain in your own words**

that are important because they help organize the code and make it easier to understand and we can reuse the same code

2. **A program has several types of employees. All employees have a name and an ID, but managers and programmers have different responsibilities. Which OOP principle could you use to represent this relationship? Explain why.**

i would use inheritance because all have the same or things in common

3. **A system has different types of vehicles, such as cars, motorcycles, and trucks. They can all perform a common action, but each type performs it differently. Which OOP principle would be useful in this situation? Explain your answer.**

i'll use Polymorphism because the vehicles can do the same actions but each vehicle it's depends on the brand and can do it in a different way.

4. **A class contains information that should not be directly accessible or modified from outside the class. Which OOP principle would help protect this information? Explain why.**

well here is encapsulation because it protects the information so i can change the information and make it more private

5. **A software system contains many complex operations, but the user only needs to interact with a few simple features. Which OOP principle could help hide the unnecessary complexity? Explain your reasoning.**

and here we use abstraction bc makes the system more simple to use

Presenta

Ing. Pedro Said Vara Chacón
Titular de la Materia

Autoriza

Ing. Josué Sebastián Aviña González
Director Académico